

# Manan Joshi

Pathankot, Punjab  
Portfolio Website

☎ +916283453432 ✉ [joshimanan265@gmail.com](mailto:joshimanan265@gmail.com) [in](#) [Linkedin](#) [G](#) [Github](#) [Codeforces](#) [LeetCode](#)  
[GeeksforGeeks](#) [CodeChef](#)

## EDUCATION

Dr. B.R. Ambedkar National Institute of Technology , Jalandhar

2023 - Current

B.Tech - Information Technology - CGPA - 8.28

Jalandhar , Punjab

## EXPERIENCE

C++ Programming Intern | CodeAlpha (Remote)

May 2025 - June 2025

- Accelerated the delivery of 4 C++ projects by abstracting complex logic into reusable OOP classes, reducing code redundancy by 40%.
- Programmed a file-based authentication mechanism for 100+ users, utilizing Regex for strict input validation to prevent data corruption.
- Engineered a secure, modular Banking System with transactions and record tracking for 100+ users.
- Refined modular codebase in a remote setup, cutting development errors by 25% via early-stage bug tracking.

## PROJECTS

FinTrack : Personal Finance Dashboard (MERN) [↗](#)

Feb 2026

- Manual financial tracking lacks predictive forecasting and relies on rigid, error-prone data entry, creating friction in personal expense management.
- Architected a dedicated Express.js backend to decouple API routing, integrating Mindee AI to asynchronously parse unstructured receipt payloads and map them into strict Mongoose schemas.
- Engineered a computationally heavy state engine utilizing a custom 3-month moving average algorithm to analyze historical ledger data, forecast baseline expenses, and dynamically execute 50-30-20 budget reallocations.
- Secured all critical REST API endpoints utilizing stateless JWT payloads, ensuring zero-exposure environments for sensitive user ledger data while fully automating the financial tracking pipeline.

Tech Stack : React.js, Node.js, Express.js, MongoDB, Tailwind CSS, Mindee AI, JWT

Talk-A-Tive: Real-Time Chat App (MERN, Socket.io) [↗](#)

Apr 2025

- Standard HTTP request-response cycles create unacceptable latency and server overhead for bidirectional, continuous messaging systems.
- Orchestrated a persistent, full-duplex communication layer utilizing Socket.io to manage concurrent real-time user connections and dynamic room states.
- Designed a robust middleware pipeline in Express.js, fortifying all API endpoints with JWT for stateless session verification and Bcrypt for cryptographic password hashing.
- Eliminated polling overhead to achieve reliable sub-100ms message delivery latency, while securing 100% of user routes against unauthorized access and boosting mobile retention by 35%.

Tech Stack : MongoDB, Express.js, React.js, Node.js, JWT, Socket.io, Tailwind CSS

## TECHNICAL SKILLS

- Languages : C++, C, Python, JavaScript, SQL, HTML/CSS
- Libraries/Frameworks : React.js, Node.js, Express.js, Next.js, Tailwind CSS, Mongoose
- Tools : Git/GitHub, VS Code, Postman, MongoDB Compass, REST API, Figma, JWT Authentication, Socket.io
- Coursework : Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks

## ACHIEVEMENTS

- National Engineering Competitor: Advanced as a National Semi-Finalist in Flipkart GRID 7.0 (Software Development Track) and ranked Top 1500 nationally in Google The Big Code 2026 out of thousands of participants.
- Applied AI & Advanced Logic: Certified in Google & Kaggle AI Agents for architecting autonomous workflows, and mastered 17 advanced graph theory paradigms under a Codeforces Master.
- Algorithmic Problem Solving: Attained LeetCode Guardian (2250+) | CodeChef 4-Star (1800+) | Codeforces Specialist (1400+) | Achieved Institute Rank 3 on GeeksforGeeks by mastering core data structures.